

DDR command-bus truth tables — DDR1 / DDR2 / DDR3 / DDR4

Command ports only. **Structural difference vs DDR5:** DDR1–4 decode the command from *dedicated* pins (CS_n, RAS_n, CAS_n, WE_n; DDR4 adds ACT_n) with bank/row/col on *separate* address pins — there is no encoded 2-cycle CA bus. All commands are **1 CK** (SDR command). DDR5 = the encoded CA bus (see [opcode_rom.pdf](#)). green =fixed command bit; yellow =address field. H=1, L=0, x=don't care.
Verified against JESD79F (DDR1 TT-3), JESD79-2F (Table 13), JESD79-3F (p47 cmd TT), JESD79-4 (Table on p32).

1 DDR1 / DDR2 / DDR3 (shared RAS/CAS/WE encoding)

command	CS_n	RAS_n	CAS_n	WE_n	BA	A10	note
Deselect (DES)	H	x	x	x	x	x	CS high: no command latched.
No-Op (NOP)	L	H	H	H	x	x	idle, holds state.
Activate (ACT)	L	L	H	H	bank	row	opens row; row on the A pins, bank on BA.
Read (RD)	L	H	L	H	bank	AP	column on A; A10 = auto-precharge.
Write (WR)	L	H	L	L	bank	AP	column on A; A10 = auto-precharge.
Precharge (PRE)	L	L	H	L	bank	all/one	A10=1 all banks, A10=0 the one BA.
Auto-Refresh (REF)	L	L	L	H	x	x	CKE high both edges.
Self-Refresh entry (SRE)	L	L	L	H	x	x	same code as REF but CKE H→L.
Mode-Reg Set (MRS)	L	L	L	L	MR sel	opcode	BA selects MR/EMR; A = opcode.
ZQ Cal (DDR3 only)	L	H	H	L	x	long/short	ZQCL (A10=1) / ZQCS (A10=0).
Power-Down entry (PDE)	L	H	H	H	x	x	NOP/DES with CKE H→L.
PD / SR exit (PDX/SRX)	CKE L→H (command pins x); DRAM resumes after tXP / tXS.						

DDR1: 4 banks (BA0–1), no ODT, no reset_n, no ZQ. DDR2: adds ODT, up to 8 banks (BA0–2), extra EMRs. DDR3: 8 banks, adds reset_n, ZQ calibration, dynamic ODT, write-leveling. Command decode is identical across the three — only the surrounding pins/features grow.

2 DDR4 (adds ACT_n; RAS/CAS/WE repurposed as A16/A15/A14)

When ACT_n=L the command is ACTIVATE and the RAS/CAS/WE pins carry the high row bits A16/A15/A14. When ACT_n=H those same pins decode every other command (as RAS_n/CAS_n/WE_n).

command	CS_n	ACT_n	A16/RAS	A15/CAS	A14/WE	BG,BA	A10	
Deselect (DES)	H	x	x	x	x	x	x	
Activate (ACT)	L	L	A16	A15	A14	grp,bank	row	
No-Op (NOP)	L	H	H	H	H	x	x	
Read (RD)	L	H	H	L	H	grp,bank	AP	
Write (WR)	L	H	H	L	L	grp,bank	AP	
Precharge (PRE)	L	H	L	H	L	grp,bank	all/one	
Auto-Refresh (REF)	L	H	L	L	H	x	x	
Mode-Reg Set (MRS)	L	H	L	L	L	MR sel	opcode	
ZQ Cal (ZQCL/ZQCS)	L	H	H	H	L	x	long/short	
SRE / PDE	L	H	x	x	x	x	x	

bank. RD/WR: column on the A pins, A10 = auto-precharge. PRE: A10=1 all banks / 0 the one BG,BA. MRS: BG,BA select MR0–MR6, A = opcode. ZQ: A10 = long/short. SRE/PDE = REF-code / NOP with CKE H→L; PDX/SRX = CKE L→H. DDR4: up to 16 banks = 4 bank groups (BG0–1) × 4 banks (BA0–1); adds ACT_n, PAR (CA parity), CID (3DS stacking), DBI, CA gear-down, tCCD_L/S. Command still 1 CK.

3 Per-version command-port differences

ver	banks	BG	cmd decode	ODT	reset_n	ZQ	ACT_n	CA PAR	DFI cmd ports
DDR1	4	-	RAS/CAS/WE	no	no	no	no	no	pre-DFI (direct)
DDR2	4-8	-	RAS/CAS/WE	yes	no	no	no	no	ras cas we, cs cke odt, addr bank
DDR3	8	-	RAS/CAS/WE	dyn	yes	yes	no	opt	+ reset_n
DDR4	16	4 (BG0-1)	ACT_n + RAS/CAS/WE	yes	yes	yes	yes	yes	act_n + address(A16-14) + bg + parity
DDR5	32	8 (3b)	encoded CA0-13 (2-cyc)	CA/CS	yes	MPC	n/a	yes	address(CA) + cs + cke + odt + parity, no ras cas we

Trend: dedicated RAS/CAS/WE pins (DDR1-3) → ACT_n overlay freeing 3 pins as address (DDR4) → fully encoded 2-cycle CA bus, no command pins at all (DDR5). Bank count ×8 across the line (4→32); bank groups appear at DDR4 (tCCD_L/S). DDR5 row = the opcode_rom.pdf table.